MIL-H-15217E
24 February 1983
SUPERSEDING
MIL-H-15217D
17 January 1975

MILITARY SPECIFICATION

HOSE, RUBBER, SANDBLAST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rubber hose with antistatic features for wet and dry sandblasting.

1.2 Classification. Hose covered by this specification shall be of one type.

1.2.1 Size. Hose covered by this specification shall be of the following inside diameter (inch) - 0.75, 1.00, 1.25, 1.50, and 2.00 sizes as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.

STANDARDS

FEDERAL

FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4720

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(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D380 - Rubber Hose, Methods of Testing.
ASTM D412 - Rubber Properties in Tension, Test Methods for.
ASTM D413 - Rubber Property - Adhesion to Flexible Substrate, Test Method for.
ASTM D518 - Rubber Deterioration - Surface Cracking, Test Method for.
ASTM D573 - Rubber - Deterioration in an Air Oven, Test Method for.
ASTM D1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen), Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a hose length for first article inspection and approval (see 4.2.1 and 6.4).

3.2 Standard commercial product. The hose shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the hoses being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance as specified herein. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Design and construction. The hose shall consist of an inner tube of elastomeric material, layer or layers of reinforcement as needed to meet the

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requirements of this specification and an outer cover of elastomeric material. The hose shall conform to the requirements of table I when tested as specified in section 4.

TABLE I. Physical performance requirements.

Requirements	Nominal size in inches (inside diameter) 0.75	1.0	1.25	1.5	2.0	Test paragraph
Inside diameter, inches	0.75	1.0	1.25	1.5	2.0	4.5
Tolerance, ±	0.031	0.031	0.031	0.031	0.031	4.5
Outside diameter[1/], Inches tolerance, ±	1.5 0.031	1.875 0.063	2.187 0.063	2.375 0.063	2.937 0.063	4.5 4.5
Thickness min						
Tube	0.25	0.25	0.25	0.25	0.25	4.5
Cover	0.031	0.031	0.031	0.031	0.063	4.5
Hydrostatic burst pressure, min pounds per square inch, (psi)	500	500	500	500	500	4.6.1
Adhesion (static-mass method), min lb:						
Between cover and plies	12	12	12	12	12	4.6.2
Between plies	12	12	12	12	12	4.6.2
Between tube and plies	12	12	12	12	12	4.6.2
Properties of tube: Tensile strength, min psi	2000	2000	2000	2000	2000	4.6.3
Decrease in tensile strength, max. percent, after air oven aging	25	25	25	25	25	4.6.4
Ultimate elongation, min percent	300	300	300	300	300	4.6.3
Properties of cover: Tensile strength, min psi	1000	1000	1000	1000	1000	4.6.3
Decrease in tensile strength, max. percent, after air oven aging	25	25	25	25	25	4.6.4
Ultimate elongation min percent	250	250	250	250	250	4.6.3
Resistance to ozone	No cracks under 2X magnification (all sizes)					4.6.5
Longitudinal electrical resistance, megohms per lineal foot max.	0.5	0.5	0.5	0.5	0.5	4.6.6

1/ When commercial hose is used, manufacturers outside diameter will be acceptable.

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3.4.1 Length. Unless otherwise specified (see 6.2), the length of the hose shall be 50 feet.

3.4.2 Reinforcement. The reinforcement shall consist of sufficient plies of square woven, cord fabric, braided or spiral yarns of natural or synthetic fibers necessary to meet the requirements of this specification and suitable for the purpose intended. Where used, the square woven fabric shall be applied on a bias of 45° with edges lapped at least 0.5 inch. The reinforcement shall be evenly and firmly woven and free from unsightly defects.

3.4.3 Tube. The tube shall be made of a compound suitable for the intended service which meets the requirements of table I. The tube shall be electrically conductive.

3.4.4 Cover. The cover shall be made of a compound suitable for the intended service which meets the requirements of table I.

3.5 Marking. Each length of hose shall be marked in a color that contrasts with the color of the hose cover. The marking shall be accomplished either by inlaying a rubber or suitable material brand, or by applying a suitable composition ink, and bonding the marking onto the cover so that the marking cannot be removed except by mechanical means. The marking shall consist of the manufacturer's name or trademark, the quarter and year of manufacture and the manufacturer's marking indicating that hose is intended for grit blast service. Hose shall be marked in at least two locations per 50 feet, at approximately 25 foot intervals, or with a continuous legend.

3.6 Workmanship. The hoses shall be made and finished with a degree of uniformity and a grade of workmanship practicable in a well-equipped factory. They shall meet the requirements of this specification and ensure proper functioning of all parts of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one assembly when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the assembly meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining assemblies to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection all include the examination of 4.5, the tests of 4.6, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.4.1 and 4.4.2.

4.3 Inspection lot. All hose of the same size offered to the Government for delivery at one time shall be considered a lot for the purpose of inspection.

4.4 Sampling. A random sample of hose lengths shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.2 Sampling for tests. Tests shall be based on inspection level S-3 with an AQL of 4.0 percent defective.

4.5 Examination. Each of the sample hose lengths selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. The classification of defects in FED-STD-162 shall be used to determine and evaluate defects through visual inspection. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. Samples selected in accordance with 4.4.2 shall be subjected to the following tests in accordance with applicable methods indicated in table II and paragraphs 4.6.1 through 4.6.6. When tested in accordance with these procedures, failure to meet the requirements of table I shall constitute cause for rejection (see 3.4).

TABLE II. Tests.

Test	Applicable test method
Hydrostatic burst test (500 psi)	ASTM D380
Adhesion (static mass method)	ASTM D413
Tensile strength and ultimate elongation (tube and cover)	ASTM D412
Accelerated aging (air oven)	ASTM D573
Ozone resistance	ASTM D518 and ASTM D1149
Longitudinal electrical resistance	Para. 4.6.6

4.6.1 Hydrostatic burst test. Hose samples shall be subjected to the hydrostatic straight bursting test of ASTM D380. Water shall be used as the test media.

4.6.2 Adhesion tests. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined in accordance with ASTM D413, with ring specimens, using the static mass method except that the required minimum weight (12 pounds) shall be supported by the test specimen for 1 minute without causing a separation of more than 1 inch.

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4.6.3 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation shall be determined in accordance with ASTM D412.

4.6.4 Accelerated aging test. Hose samples shall be subjected to accelerated aging in accordance with ASTM D573, except that the time for aging shall be 70 hours at a temperature of 158° Fahrenheit (F). After aging, the tensile strength shall not have changed more than specified in table I.

4.6.5 Ozone resistance test. Specimens of the cover, prepared as described in Procedure A of ASTM D518, shall be mounted in a 20 percent elongated position and tested in accordance with ASTM D1149. After conditioning for 48 hours in an ozone-free atmosphere, the mounted specimen shall be exposed for 72 hours at 104 $\pm$3.6°F to an ozone concentration of 50 $\pm$5 parts per one hundred million by volume.

4.6.6 Longitudinal electrical resistance. The outside of the hose shall be thoroughly cleaned to remove any conducting powders. The hose ends shall be squared off. The inside surface of the hose shall be buffed with sandpaper for a distance of 3 inches from each end. A tight-fitting nipple shall be inserted into each end for a distance of 2 inches and clamped. The hose shall be suspended horizontally above the ground by any means which has infinite electrical resistance to the ground. The resistance between the nipples shall be measured with a standardized megohmmeter. The hoses shall be at a temperature of 75 $\pm$5°F.

4.7 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-775 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. Hose covered by this specification is intended for use to convey sand, metal shot, or grit entrained with water or compressed air to a control nozzle for cleaning castings and other metal surfaces.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If a first article is required (see 3.1, 4.2.1, and 6.4).
c. If the hose length shall be other than 50 feet (see 3.4.1).
d. Size of hose required (see 1.2.1 and table I).
e. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete assembly, or it may be a standard production item from the contractor's currect inventory as specified in 4.2.1. The contracting officer should include specific instruction in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Navy - YD
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Review activity:

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Preparing activity:

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